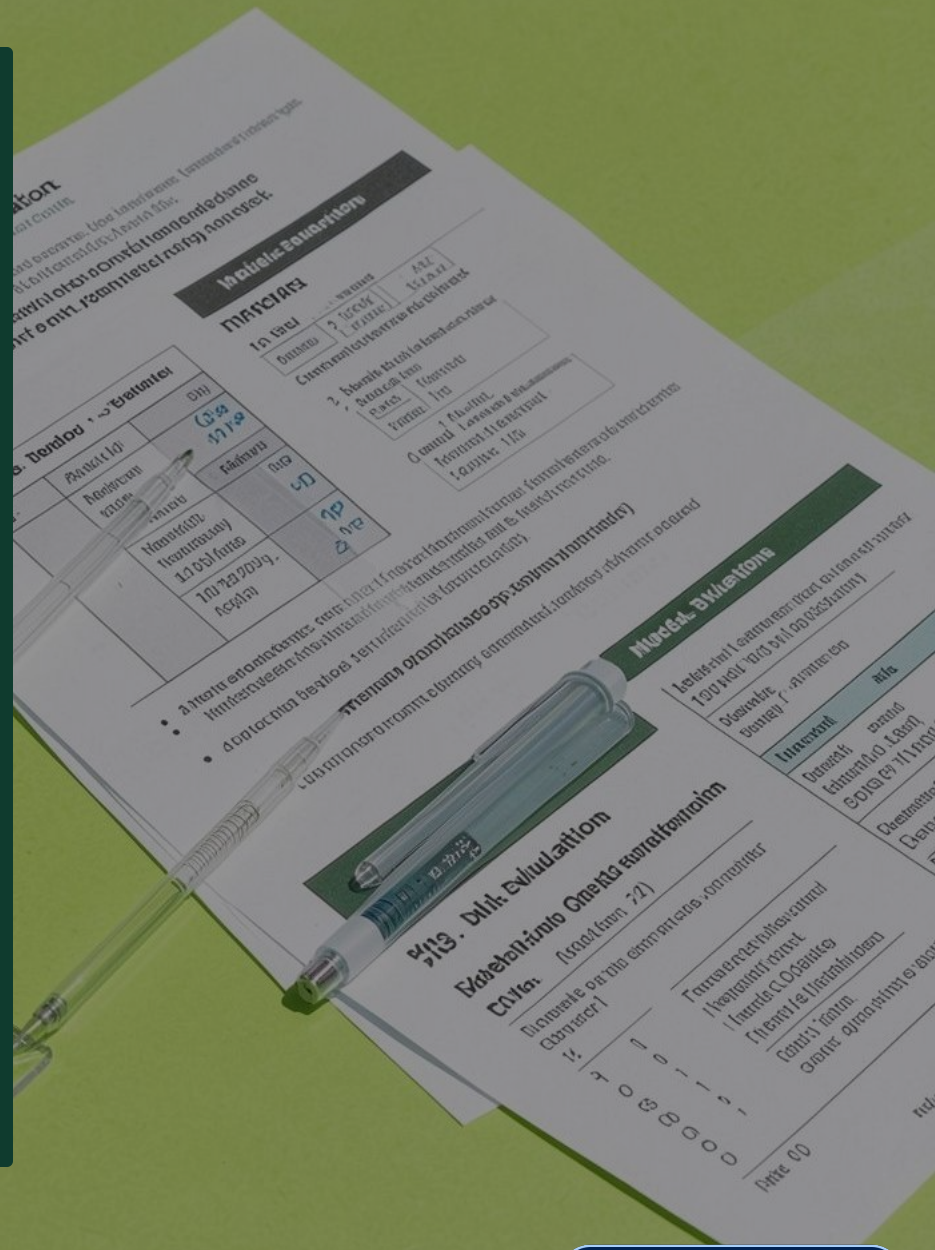
MACHINE-LEARNING EVALUATION

Performance Analysis and Merit of a Machine-Learning System

Name: Guru Charan M
Reg. No.: 192524434
Course: Machine Learning

From predictive quality to dependable deployment



Made with GAMMA

THE EVALUATION QUESTION

A high score is only the start.

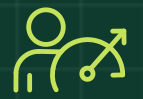
A strong model is more than a high accuracy score

A credible evaluation asks what the model gets right, what it gets wrong, and whether it can operate reliably.



Predictive quality

Evaluate the developed system with quantitative metrics and a fair comparison against baseline methods.



Operational efficiency

Consider the time, memory, compute, and effort required to produce each prediction.



Dependable behavior

Judge whether performance remains stable across data, classes, and changing conditions.

Made with GAMMA

01 · METRICS

Start with the right evaluation metrics

Each metric answers a different question about the same predictions.

1. **Accuracy**
   Overall share of correct predictions.
2. **Precision**
   How often positive predictions are correct.
3. **Recall / sensitivity**
   How many actual positives the model detects.
4. **F1-score**
   Balance between precision and recall.
5. **Error rate**
   The mistakes that remain after prediction.

Made with GAMMA

The confusion matrix reveals what accuracy hides

Read each cell as a count of model decisions, not as an abstract score.

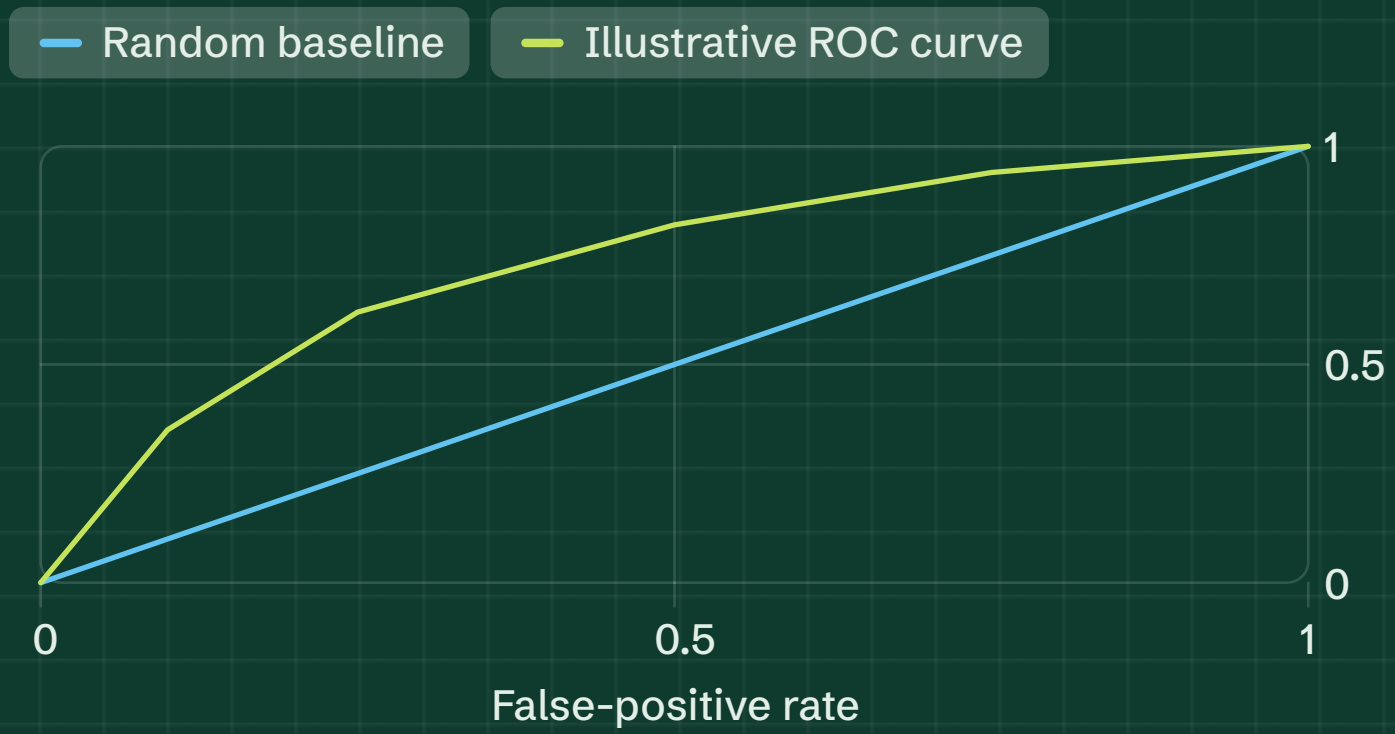
Use the matrix to identify the costliest error type for the application.

Merit changes with the error cost. In a safety-critical task, a false negative may matter more than a false positive.

Connect the counts to precision, recall, specificity, and error analysis.

Actual class	Predicted positive	Predicted negative
Positive	**True positive** Positive case detected.	**False negative** Positive case missed.
Negative	**False positive** Negative case flagged.	**True negative** Negative case rejected.

Made with GAMMA



Illustrative values only · project ROC points and AUC were not supplied.

03 · THRESHOLDS

ROC analysis shows performance across thresholds

Plot the true-positive rate against the false-positive rate as the decision threshold changes.

OPERATING REGION

The curve and AUC offer a comparative view, not a replacement for application-specific metrics.

The best operating point depends on the cost of missed positives versus false alarms.

Made with GAMMA

04 · FAIR COMPARISON

Compare the proposed system with a meaningful baseline

Same data split, task, and evaluation rules; make the improvement visible.

Measure	Proposed system	Baseline A	Baseline B
Accuracy	TBD	TBD	TBD
Sensitivity	TBD	TBD	TBD
Precision	TBD	TBD	TBD
F1-score	TBD	TBD	TBD
Error rate	TBD	TBD	TBD
Inference time	TBD	TBD	TBD
Resource use	TBD	TBD	TBD

Placeholder view · replace TBD with measured results. State the direction: higher quality, fewer errors, faster execution, or better resource efficiency.

Made with GAMMA

Merit includes efficiency, reliability, and scalability

Predictive quality matters, but deployment determines whether the system is useful.



Efficiency

Measure: latency, throughput, training cost, and memory or compute requirements.

`Can the system deliver predictions within the available budget?`



Reliability

Check: stability across folds, classes, datasets, or changing conditions.

`Does the result hold beyond one convenient test?`



Scalability

Ask: whether performance holds as data volume, users, or deployment load grows.

`Can the system expand without losing its value?`

Made with GAMMA

THE DECISION

Merit is a chain of evidence.

Connect the metric to the decision the system is meant to support.

1. **Quality**

   High accuracy matters only when it is supported by improved sensitivity and lower error rates.

2. **Evidence**

   Confusion matrices, ROC curves, and performance graphs make the evidence inspectable.

3. **Merit**

   The proposed system earns merit when it outperforms the baseline fairly and remains efficient, reliable, and scalable.

The conclusion should name what improved, why it matters, and what evidence supports the claim.

Made with GAMMA